1. Write a Hadoop command to implement file management tasks in HDFS environment.Register No.Aim &Procedure (20)Commands/
Program(40)Execution & Output(30)Viva-Voce(10)Total(100) procedure
should be atleast 7 points and programs and commands are for windows not
linux or mac os

4:19 am Experiment 1: HDFS File Management Tasks (Windows)
Aim

To implement basic file management tasks in the Hadoop Distributed File System (HDFS)

environment using a Windows operating system. Procedure

1. Open the Windows Command Prompt with Administrator privileges. 2. Navigate to your Hadoop installation directory bin path (e.g., C:\hadoop\sbin). 3. Start the core Hadoop storage daemons by running the Windows batch script start-dfs.cmd. 4. Verify that the NameNode and DataNode processes are running successfully in separate

command windows. 5. Use the Windows-compatible file system shell command format hdfs dfs -<command> to interact
with storage paths. 6. Run basic system verification checks to list files, display file properties, and view disk usage
information. 7. Confirm file transfer operations by moving sample files from your Windows local file path directly
into the HDFS environment. Commands
cmd

:: 1. Navigate to the Hadoop distribution sbin directory

cd C:\hadoop\sbin

:: 2. Start HDFS services for Windows execution

start-dfs.cmd

:: 3. List the root directory structure inside HDFS
hdfs dfs -ls /

:: 4. View the disk space utilization of a directory path in human-readable format hdfs dfs -du -h /

:: 5. Download an HDFS file back down to your local Windows Downloads folder

hdfs dfs -get /hdfs_dir/sample.txt C:\Users\Public\Downloads\

:: 6. Display the file content streams directly inside the Command Prompt window hdfs dfs -cat /hdfs_dir/sample.txt

:: 7. Check the operational status of your Windows-based cluster nodes
hdfs dfsadmin -report

Use code with caution. Sample Output
text
C:\hadoop\sbin>hdfs dfs -ls /
Found 2 items
drwxr-xr-x - Administrator supergroup 0 2026-06-03 10:00 /hdfs_dir drwxr-xr-x - Administrator supergroup 0 2026-06-03 10:15 /tmp

C:\hadoop\sbin>hdfs dfs -cat /hdfs_dir/sample.txt
Hello Hadoop on Windows! This is a test file execution.